



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
Office of the Vice President for Campuses
PARANAQUE CITY CAMPUS



“Kawayan AI: The Design and Implementation of an Intelligent Content Generation and Scheduling Platform for Enhanced Philippine Business Visibility”

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RESEARCH COLLOQUIUM 2026



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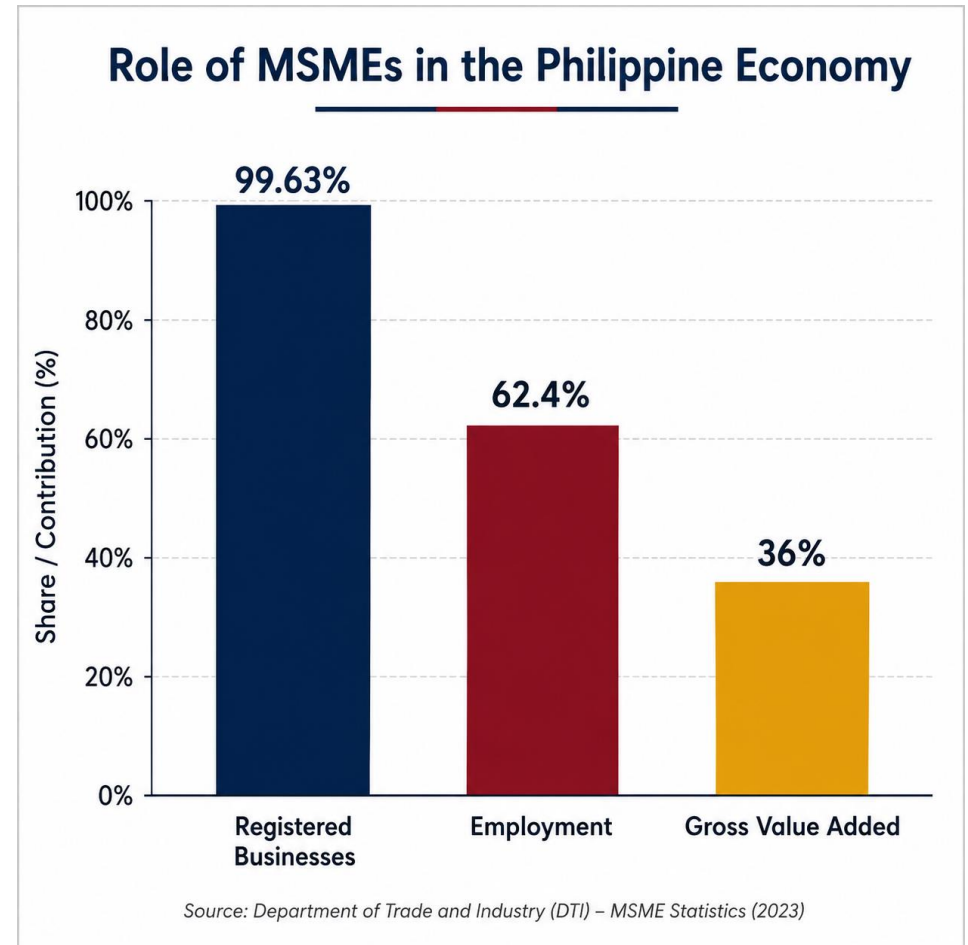
Chapter 1

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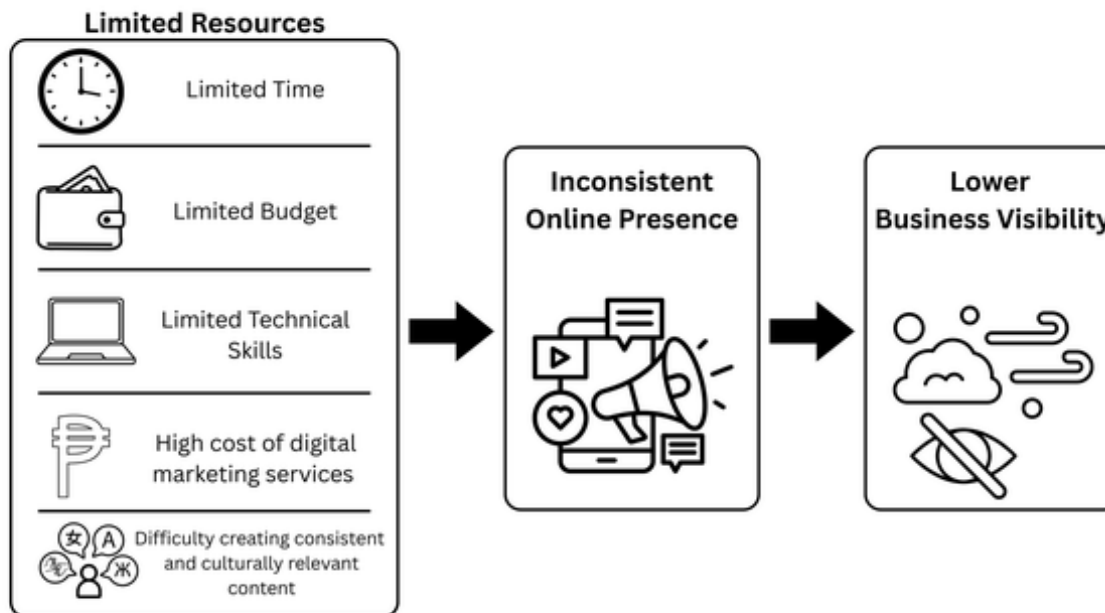
Introduction

- MSMEs are vital to the Philippine economy, accounting for 99.63% of registered businesses, 62.4% of employment, and 36% of gross value added.





Introduction



- Despite high social media usage in the Philippines, many MSMEs remain digitally marginalized due to limited time, budget, technical skills, and access to affordable digital marketing services.

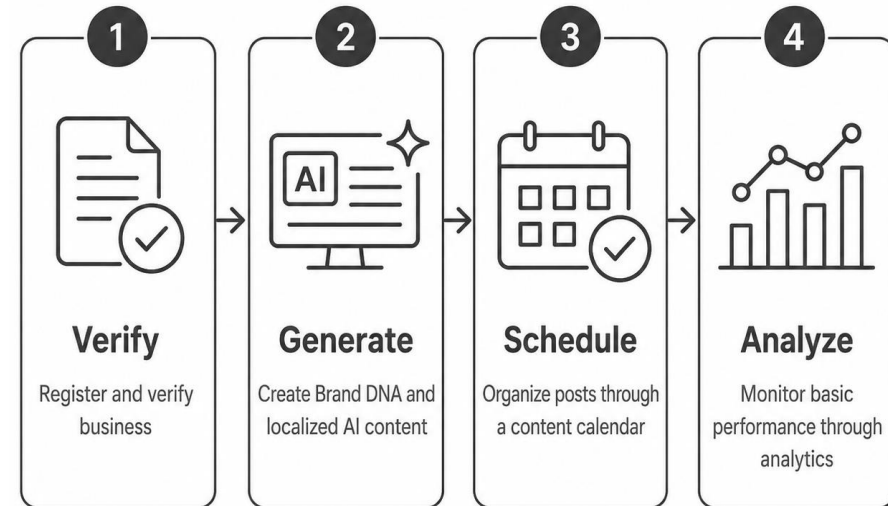


Introduction

- Kawayan AI is a web-based intelligent content generation and scheduling platform designed for Philippine MSMEs.

The system helps users:

- Register and verify their business
- Create a Brand DNA profile
- Generate localized AI-assisted content
- Organize posts through a content calendar
- Monitor basic performance through analytics





SDG ALIGNMENT

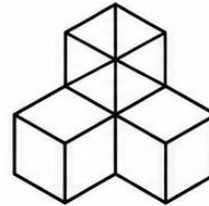
This study aligns with the Sustainable Development Goals:

8 DECENT WORK AND ECONOMIC GROWTH



Supports MSME productivity, digital competitiveness, and economic participation through AI-assisted marketing automation.

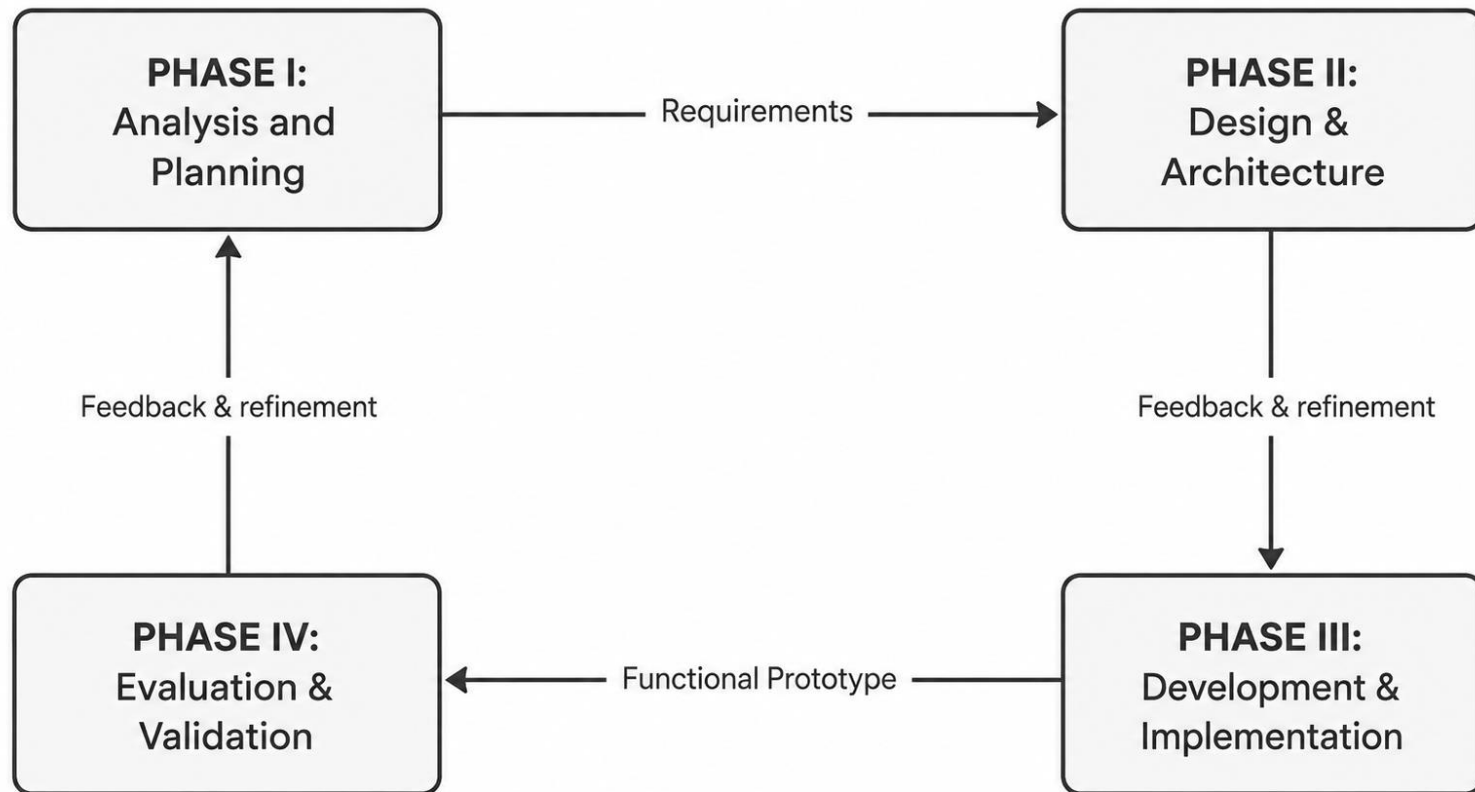
9 INDUSTRY, INNOVATION, AND INFRASTRUCTURE



Promotes localized technological innovation through culturally intelligent AI, content automation, and digital infrastructure for MSMEs.



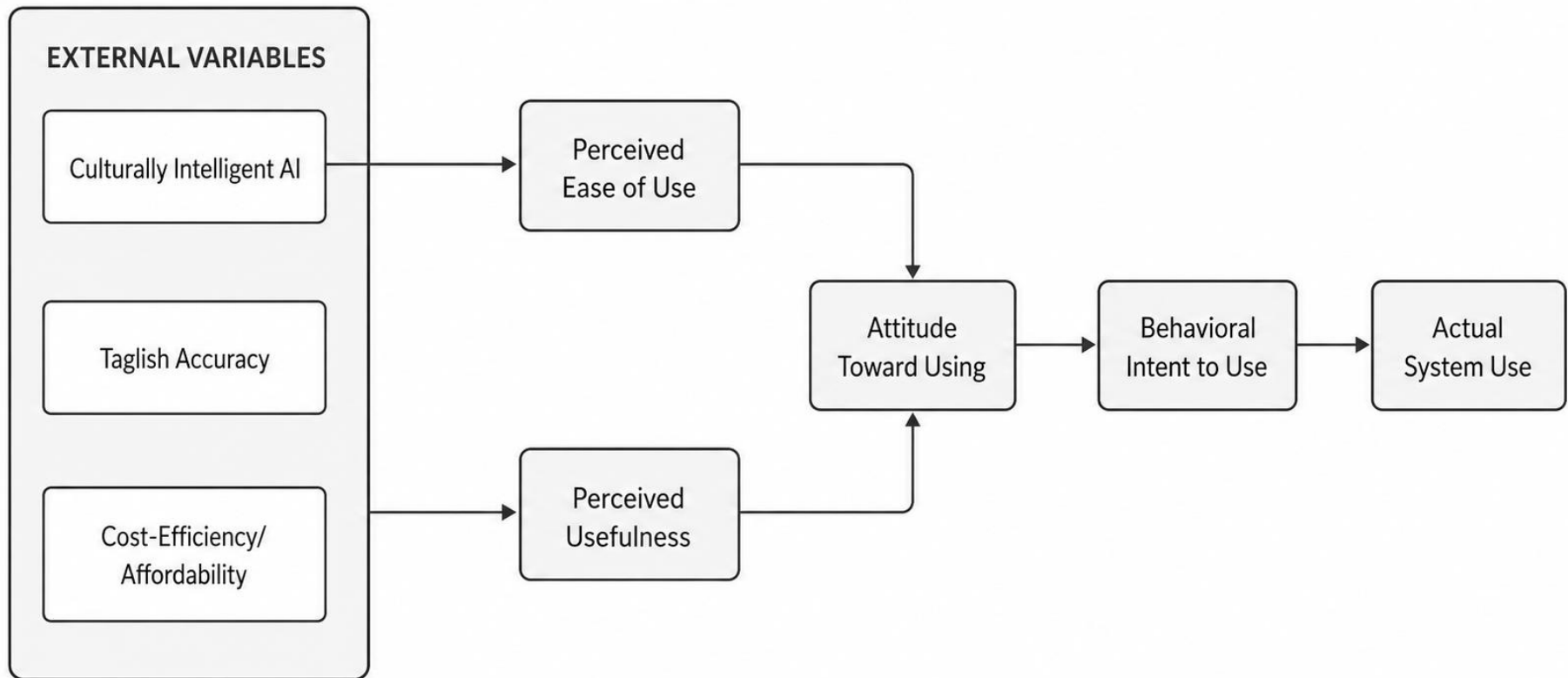
Theoretical Framework



Design and Development Research (DDR) Cycle



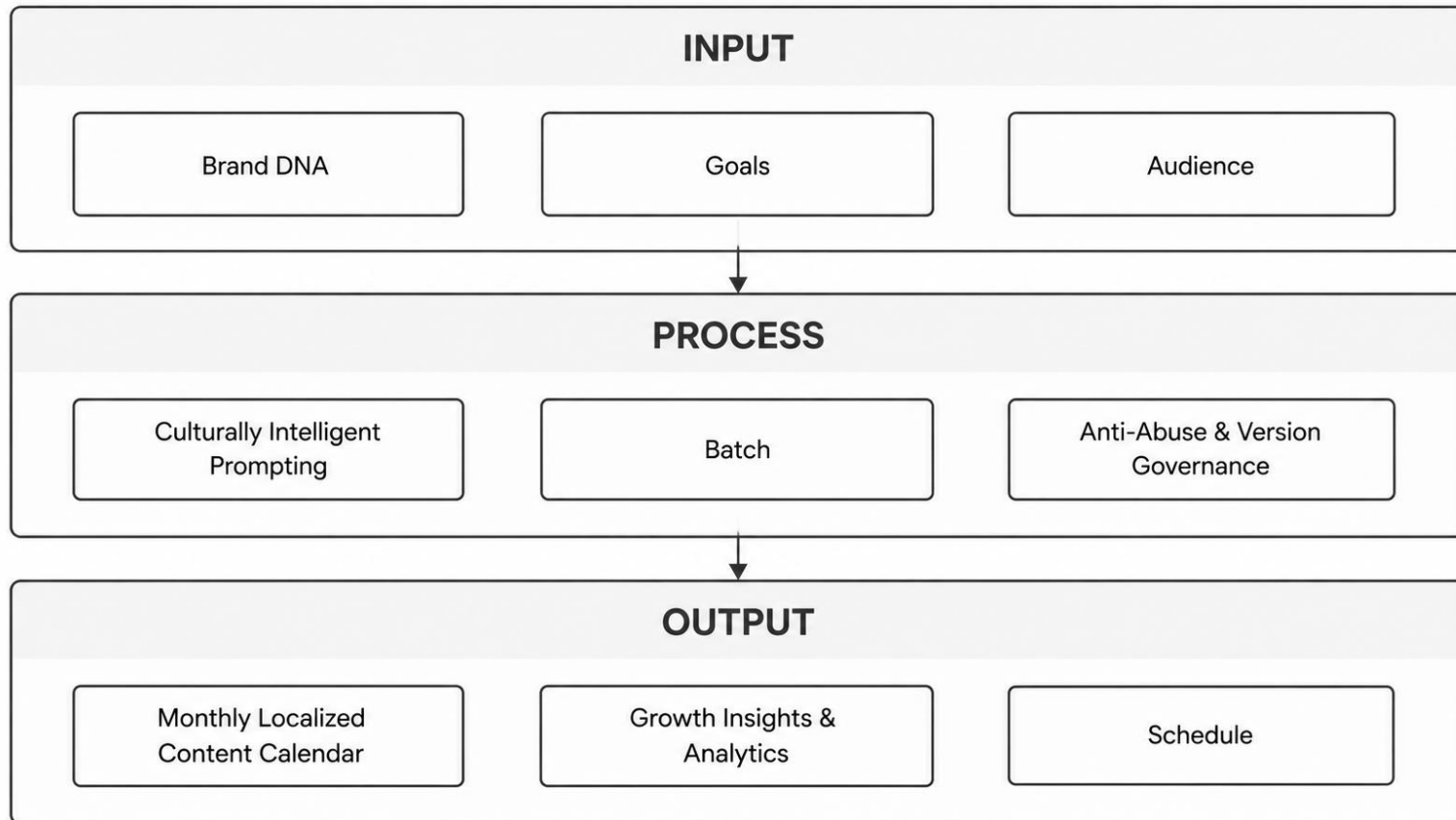
Theoretical Framework



Adapted Technology Acceptance Model (TAM)



Conceptual Framework





Statement of the Problem

The study addresses three main problems faced by MSMEs:

1. Prohibitive Operational Costs: Digital marketing services are expensive for many micro and small enterprises.
2. Lack of Cultural Localization: Generic AI tools may fail to produce natural Taglish and culturally relevant content.
3. Inefficient Time and Resource Management: MSME owners lack time and tools for consistent content planning, scheduling, and monitoring.



Research Questions

- How can Kawayan AI help MSMEs create localized marketing content faster?
- How can the platform support content scheduling and dashboard monitoring?
- How acceptable is the system in terms of usefulness and ease of use?
- How effective is the system in producing culturally relevant Filipino/Taglish content?
- How can business verification support secure and credible platform access?



Significance of the Study

- **MSMEs:** Provides affordable support for content generation, scheduling, and analytics.
- **Public Consumers:** Helps businesses produce more relatable and culturally relevant online content.
- **Local Community:** Supports local business visibility and competitiveness.
- **Future Researchers:** Serves as reference for AI, MSME digitalization, and localized marketing systems.
- **Developers and IT Students:** Provides a practical model for AI-based business solutions.



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Chapter 2

RESEARCH COLLOQUIUM 2026



Review of Related Literature

Author/s / Year	Research Focus	Major Findings	Gap / Link to Present Study
DTI (2024); Google, Temasek, & Bain & Co. (2025); DataReportal (2024)	Philippine MSMEs and the digital economy	MSMEs make up 99.63% of registered businesses, employ 62.4% of the workforce, and operate in a highly digital market with 86.75 million social media users.	Shows the need for digital tools that can help MSMEs improve online visibility and competitiveness.
Entrepreneurship.org.ph (2024); Ailleron (2024); Wazile (2024); MDPI (2024)	Digital marginalization and resource constraints	Many MSMEs remain digitally marginalized due to limited manpower, technology, time, budget, and high digital marketing costs.	Supports the need for an affordable and easy-to-use platform for MSME digital marketing.
Reyes & Fajardo (2021); Journals Index Copernicus (2023); Xu (2022)	Taglish and cultural localization	Filipino digital communication requires cultural and linguistic accuracy; awkward or foreign-sounding content may reduce consumer interest.	Supports Kawayan AI's focus on culturally intelligent prompting and natural Taglish content generation.



Review of Related Literature

Author/s / Year	Research Focus	Major Findings	Gap / Link to Present Study
ResearchGate (2023); Livermore & Adams (2025)	Perceived Brand Localness and culturally intelligent prompting	Localized and culturally authentic brands can strengthen trust, consumer connection, and purchase intention.	Establishes the theoretical basis for using localized AI-generated content to improve MSME brand credibility.
Jasper (2026); Google (2026)	AI-powered marketing platforms and frontier AI	Generative AI can support marketing content creation, planning, and batch ideation.	Existing AI tools are often generic, so Kawayan AI localizes AI content generation for Philippine MSME needs.
Xendit (2026); bundle.social API (2026)	Payment and scheduling infrastructure	Local payment gateways and scheduling APIs can support recurring billing and cross-platform content distribution.	Supports the feasibility of Kawayan AI's payment, scheduling, and platform automation features.



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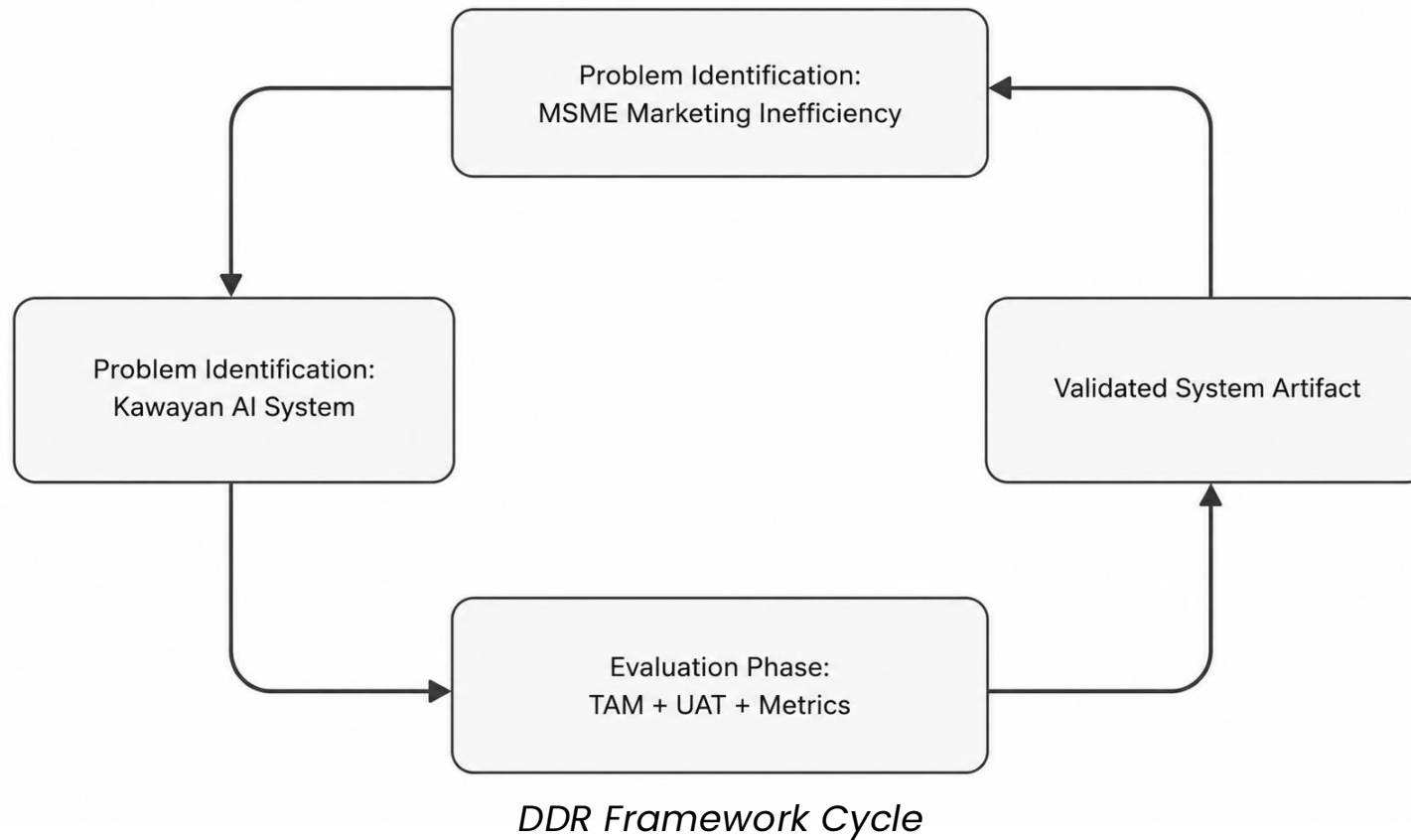


Chapter 3

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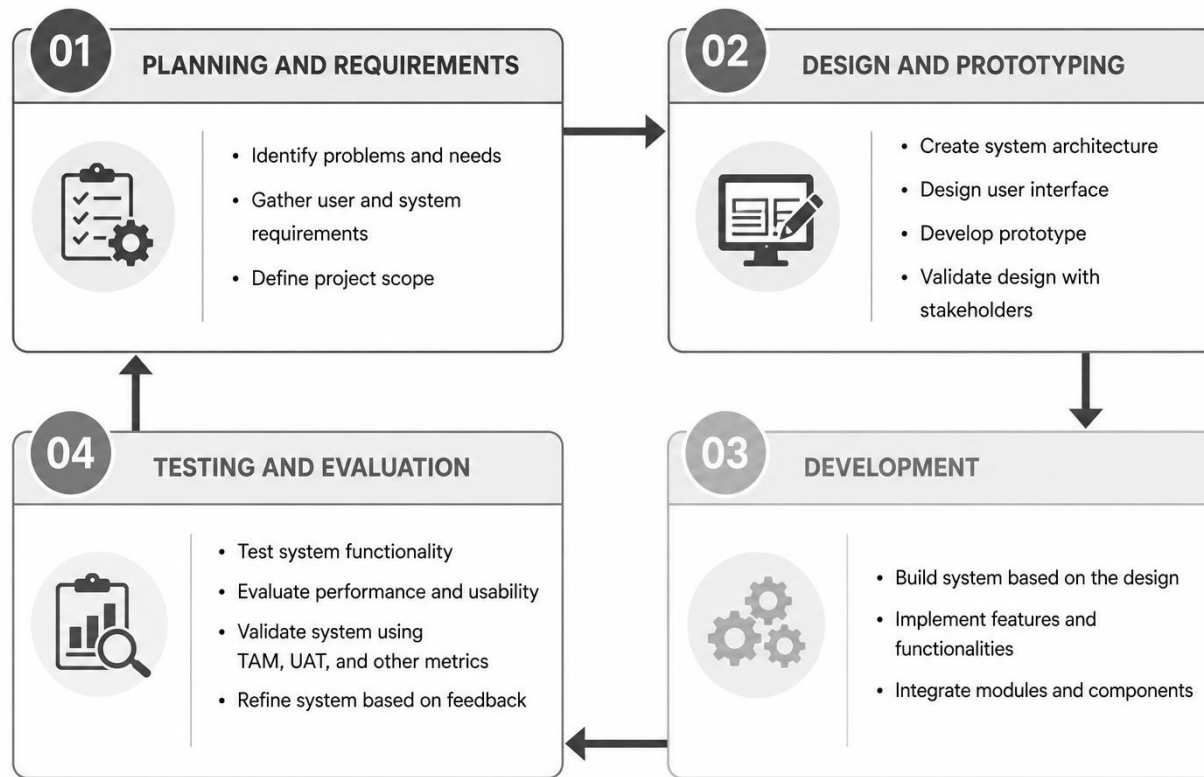


Research Design and Development Model





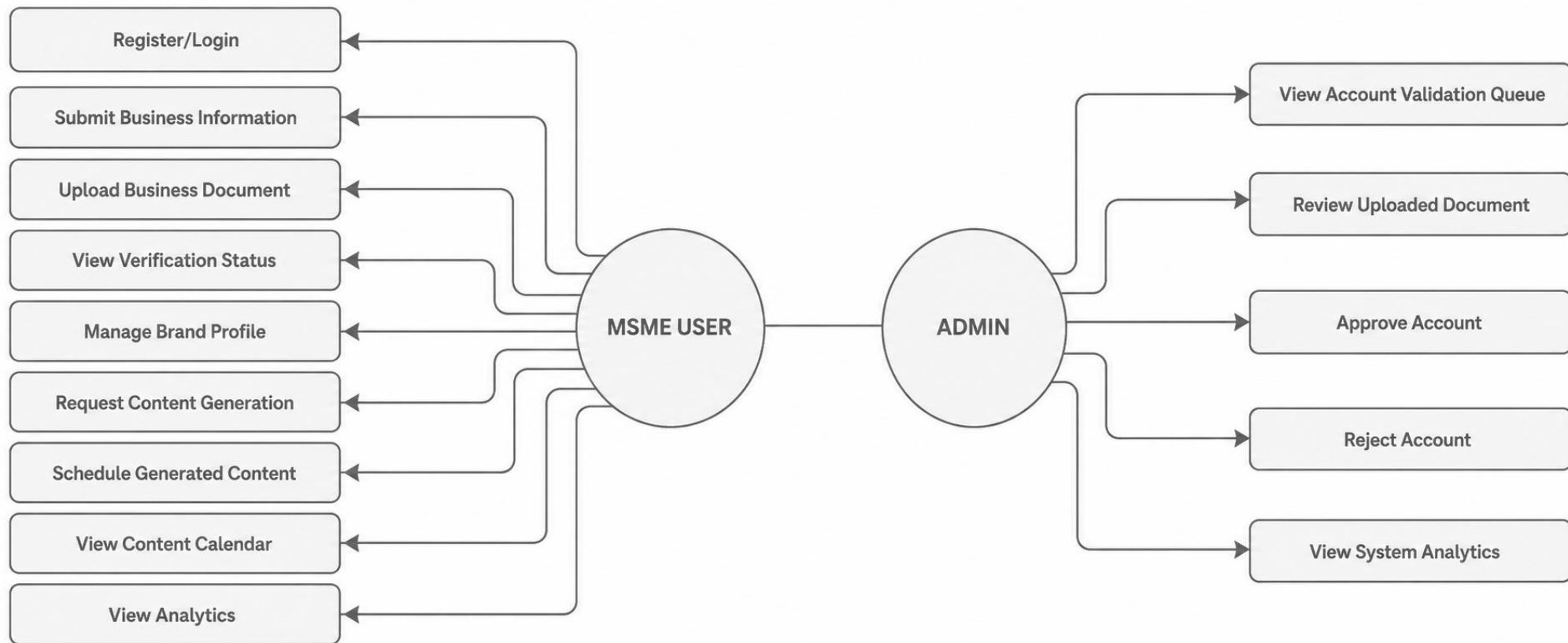
Research Design and Development Model



Iterative Prototyping Model (IPM)

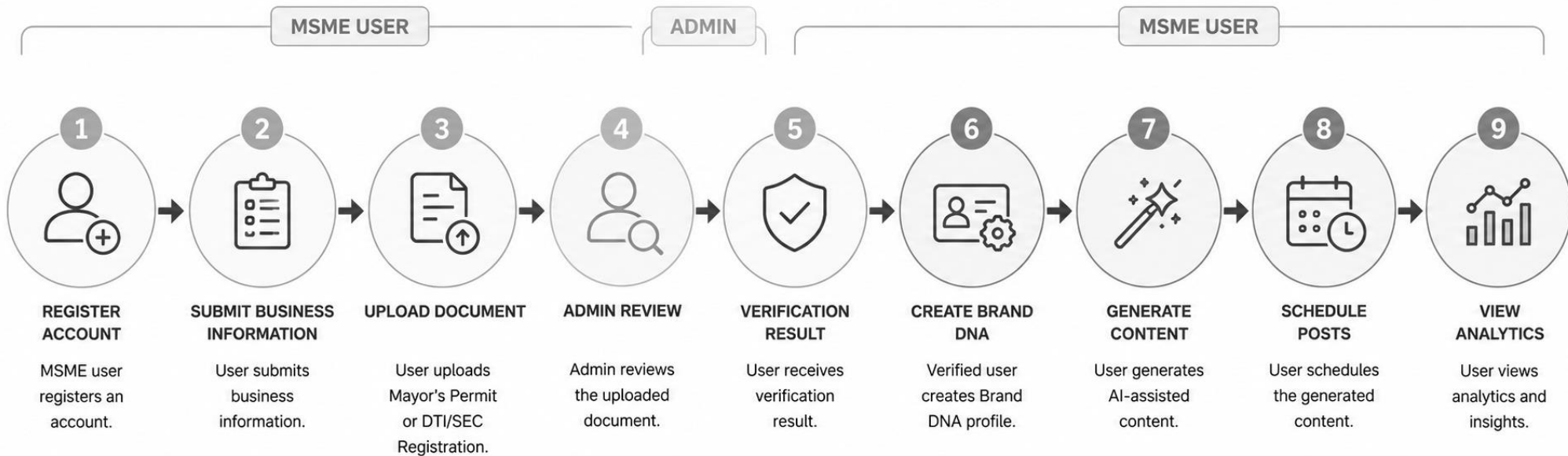


USE CASE DIAGRAM





SYSTEM PROCESS FLOW





LOCALE, PARTICIPANTS, INSTRUMENTS, AND DATA ANALYSIS

- **Locale:** Parañaque City, Philippines
- **Participants:** 40 selected MSME owners, operators, or individuals managing business social media accounts
- **Sampling Technique:** Purposive sampling
- **Research Instrument:** TAM-based survey questionnaire using a 5-point Likert scale



LOCALE, PARTICIPANTS, INSTRUMENTS, AND DATA ANALYSIS

Data Gathering:



Data Analysis:

- Frequency
- Percentage
- Weighted mean
- Likert-scale interpretation
- Cronbach's Alpha
- Review of UAT feedback